	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence C5, C6 and C7** **Continuous distributions and sample data – Version 2** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.	**Assessment criteria:** C5: Calculates and interprets the mean, variance and standard deviation of continuous distributions using standard formulas and functions. C6: Estimates interval probabilities using relative frequencies from generated samples. and compares them with theoretical probabilities. C7: Calculates sample means, variances, and standard deviations for different sample sizes and data representations.

Criteria assessment

This exam assesses C5, C6, and C7 separately (6 points each; 18 total). Each task specifies its points. Partial credit requires correct procedures; answers without procedures receive zero. Missing names or late submissions incur a 1-point deduction per criterion.**Passing requires at least 4 points per criterion**. Use unrounded intermediate values and round final statistics to 3 decimals and percentage errors to 2.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	1	–
5	–	1	–
6	–	2	–
7	–	2	–
8	–	–	1
9	–	–	1
10	–	–	2
11	–	–	2
Total	**6**	**6**	**6**

1. Two parcel-labeling stations [C5-2]

A delivery company uses two parcel-labeling stations. At station A, an automated machine labels a parcel in anywhere from 5 to 19 minutes. Within these limits, any two time windows of the same length are equally likely to contain the finishing time. Let A be its labeling time in minutes.

At station B, staff label special parcels, also taking between 5 and 19 minutes. The manager draws a smooth outline of how often different labeling times occur. The outline starts at zero at 5 minutes, rises along a straight edge to its single highest point at 10 minutes, then falls along a straight edge to zero at 19 minutes. No labeling times occur outside these limits. Let B be this labeling time in minutes.

(a) Identify the distribution for A from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

(b) Identify the distribution for B from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

2. Two custom-print studios [C5-2]

Two studios print custom posters. Each order takes between 3 and 15 hours. The owners use smooth outlines to describe how often different completion times occur over many orders; times outside these limits never occur. At studio I, longer jobs are more common: the outline is a single straight edge rising from zero at 3 hours to its highest point at 15 hours. At studio D, shorter jobs are more common: the outline is a single straight edge falling from its highest point at 3 hours to zero at 15 hours. Let I and D be the respective completion times in hours.

(a) Identify each distribution from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[2]**

3. Bookings and daily takings [C5-2]

A camera repair shop receives online bookings throughout its opening hours, averaging 10 bookings per hour. Bookings arrive independently, one at a time, at a steady pace. In a very short time window, the chance of a booking is proportional to the window's length; two or more bookings are negligible. Past waiting does not affect the chance of a booking during the next minute. Let T be the time between consecutive bookings in minutes.

On comparable working days, total revenue varies because of many small, independent influences, with none dominating. Its smooth outline has a symmetric bell shape, highest at \$3200 and tapering on both sides. Its sides change from becoming steeper to becoming flatter at \$3000 and \$3400. Treat this bell-shaped pattern as exact. Let R be daily revenue in hundreds of dollars.

(a) Identify the distribution for T from the clues and use its formulas to find the mean, variance, and standard deviation in minutes. **[1]**

(b) Identify the revenue distribution from the clues. Use its formulas to find the mean, variance, and standard deviation in dollars. Interpret the mean. **[1]**

4. Office electricity use, raw data [C6-1]

A simulated sample of daily office electricity use (tens of kilowatt-hours) is

1.3	1.7	2.0	2.2	2.6	2.9	3.3	3.5	3.8	4.2
1.6	1.9	2.3	2.7	3.1	3.4	3.6	3.9	4.4	4.6

The theoretical probability is $P(2 \le X \le 3.5) = 0.60$.

(a) Find the empirical probability of the stated interval from the given sample. **[0.5]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

5. Equipment rental times, frequency data [C6-1]

Recorded equipment rental times are summarized below.

x_i (min)	4	8	12	16	20	24
f_i	8	16	28	22	14	12

The theoretical probability is $P(8 \le X < 20) = 0.80$.

(a) Find the empirical probability of the stated interval from the given sample. **[0.5]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

6. File upload times, grouped data [C6-2]

A sample of file upload times in seconds is summarized below. Classes are left-inclusive and right-exclusive.

Class	$[2, 8)$	$[8, 14)$	$[14, 20)$	$[20, 26)$
f_i	5	9	7	3

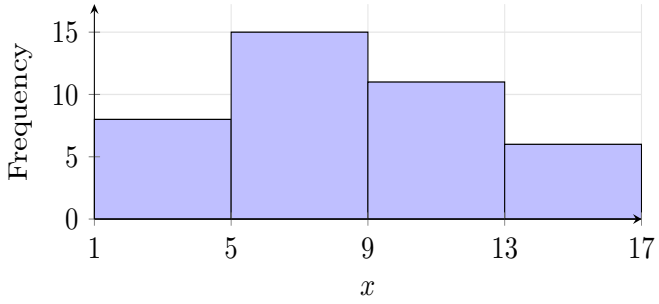
The theoretical probability is $P(8 \le X < 20) = 0.75$.

(a) Find the empirical probability of the stated interval from the given sample. **[1]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

7. Museum entrance queues, histogram data [C6-2]

The histogram is the only record of a sample of museum entrance queue times. Bar heights are frequencies; times are in minutes, with left-inclusive, right-exclusive classes.



The theoretical probability is $P(5 \le X < 13) = 0.70$.

(a) Find the empirical probability of the stated interval from the given sample. **[1]**

(b) Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

Use the sample-variance denominator $n-1$ throughout these four problems. For each statistic, calculate absolute error $E = |\text{sample value} - \text{theoretical value}|$ and percentage error $100E/|\text{theoretical value}|$. Use unrounded values in all error calculations. For grouped data and histograms, use class midpoints to estimate the statistics.

8. Chamber temperatures, raw data [C7-1]

Five chamber temperature readings in degrees Celsius are $14, 17, 18, 19, 22$. The theoretical model has $\mu = 19$ degrees Celsius and $\sigma^2 = 6$ square degrees Celsius.

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

9. Parcels dispatched, frequency data [C7-1]

The numbers of parcels dispatched in nine short collection periods are summarized below. The theoretical model has $\mu = 7$ parcels and $\sigma^2 = 9$ parcels^2.

x_i	3	7	11
f_i	2	4	3

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

10. Walking times, grouped data [C7-2]

Walking times in minutes are grouped using left-inclusive, right-exclusive classes. The theoretical mean is 23 minutes and variance is 30 minutes^2.

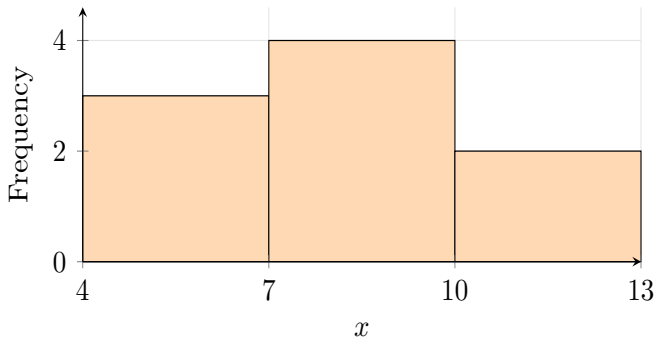
Time (min)	$[12, 18)$	$[18, 24)$	$[24, 30)$
f_i	2	3	4

(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[1]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**

11. Luggage masses, histogram data [C7-2]

A histogram gives a sample of luggage masses in kilograms. Bar heights are frequencies; classes are left-inclusive and right-exclusive. The model has $\mu = 9$ kg and $\sigma^2 = 7$ kg^2.



(a) Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[1]**

(b) Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**